Year 11 Set 2 · Higher · grade 5–6 · Lesson 2

Quadratics: factorising & solving

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Today I will learn to...

Goals

1. Factorise and solve $x^2 + bx + c = 0$

2. Form and solve quadratic equations from problems

Do now: retrieval practice

4 mins

1. Solve $x + y = 8$, $x - y = 2$

   $x = 5, y = 3$

2. Expand $(x + 4)(x - 3)$

   $x^2 + x - 12$

3. Work out $\sqrt{144}$

   12

4. HCF of $12x$ and $18x^2$

   $6x$

Objective 1

Factorise and solve $x^2 + bx + c = 0$

Key idea

Learn

!

Find two numbers that multiply to c and add to b. Set each bracket to 0. Look out for $x^2 - a^2 = (x - a)(x + a)$.

Objective 1: Factorise and solve $x^2 + bx + c = 0$

I do: example 1

I do

$x^2 + 7x + 12 = 0$

1. Find two numbers that multiply to 12 and add to 7
2. 3 and 4
3. $(x + 3)(x + 4) = 0$
4. $x = -3$ or $x = -4$

I do: example 2

I do

$x^2 - 2x - 15 = 0$

1. Find two numbers that multiply to −15 and add to −2
2. −5 and 3
3. $(x - 5)(x + 3) = 0$
4. $x = 5$ or $x = -3$

I do: example 3

I do

$x^2 - 49 = 0$

1. Difference of two squares: $a^2 - b^2 = (a - b)(a + b)$
2. $x^2 - 49 = x^2 - 7^2$
3. $(x - 7)(x + 7) = 0$
4. $x = 7$ or $x = -7$

Mini whiteboards

We do

1. Solve $x^2 + x - 20 = 0$

Answer: $x = -5$ or 4

2. Solve $x^2 - 9x = 0$

Answer: $x = 0$ or 9

3. Solve $x^2 + 7x + 12 = 0$

Answer: $x = -3$ or -4

4. Solve $x^2 - 16 = 0$

Answer: $x = 4$ or -4

You do: choose your challenge

You do

Bronze

Solve $x^2 + 6x + 8 = 0$

Silver

Solve $x^2 - 3x - 10 = 0$

Gold

Solve $x^2 = 4x + 21$

Platinum

A rectangle is $(x + 3)$ by $(x - 2)$ and has area 36. Find x and the perimeter.

You do: answers

Check

Bronze

$x = -2$ or -4

Silver

$x = 5$ or -2

Gold

$x^2 - 4x - 21 = 0$, $x = 7$ or -3

Platinum

$x^2 + x - 42 = 0$, $x = 6$. Perimeter = 26

If you finish early

Extra

Bronze

Solve $x^2 + 5x + 6 = 0$

$x = -2$ or -3

Silver

Solve $x^2 - x - 12 = 0$

$x = 4$ or -3

Gold

Solve $x^2 + 2x = 24$

$x = 4$ or -6

Platinum

A square has side $(x + 2)$ cm and area 49 cm^2. Find x.

$(x + 2)^2 = 49$, so $x = 5$

Exam question 1

3 marks

Solve $x^2 - 8x + 15 = 0$

Space for working

Mark scheme: $(x - 3)(x - 5) = 0$, $x = 3$ or 5

Exam question 2

2 marks

Factorise fully $2x^2 - 18$

Space for working

Mark scheme: $2(x - 3)(x + 3)$

Objective 1 achieved

Factorise and solve $x^2 + bx + c = 0$



Objective 2

Form and solve quadratic equations from problems

✓ Achieved: Factorise and solve $x^2 + bx + c = 0$

Key idea

Learn

!

Write an expression, set it equal to the value, rearrange to = 0, factorise and solve. Reject answers that make no sense, like a negative length.

Objective 2: Form and solve quadratic equations from problems

I do: example 1

I do

A rectangle has width x cm and length (x + 4) cm. Its area is 45 cm². Find x.

1. $x(x + 4) = 45$
2. $x^2 + 4x - 45 = 0$
3. $(x + 9)(x - 5) = 0$
4. $x = 5$ (a length can't be negative)

I do: example 2

I do

A number squared plus 3 times the number is 40. Find the number.

1. $n^2 + 3n - 40 = 0$
2. $(n + 8)(n - 5) = 0$
3. $n = 5$ or $n = -8$

I do: example 3

I do

A square has side (x + 2) cm and area 49 cm². Find x.

1. $(x + 2)^2 = 49$
2. $x + 2 = 7$ (side is positive)
3. $x = 5$

Mini whiteboards

We do

1. $x(x + 3) = 28$. Find the positive value of x.

Answer: $x = 4$

2. $n^2 - n = 30$. Find n.

Answer: $n = 6$ or $n = -5$

3. $x(x - 2) = 35$. Positive x?

Answer: $x = 7$

4. Two consecutive integers multiply to 56. Find them.

Answer: 7 and 8 (or −8 and −7)

You do: choose your challenge

You do

Bronze

$x(x + 2) = 15$. Find the positive value of x.

Silver

A rectangle is x cm by $(x + 5)$ cm with area 50 cm^2. Find x.

Gold

Two consecutive whole numbers multiply to 72. Find them.

Platinum

A lawn is 2 m longer than it is wide. A 1 m path goes all round it. Lawn and path cover 63 m^2. Find the lawn's width.

You do: answers

Check

Bronze

$x = 3$

Silver

$(x + 10)(x - 5) = 0$, $x = 5$

Gold

8 and 9 (or −9 and −8)

Platinum

$(w + 2)(w + 4) = 63$, $w^2 + 6w - 55 = 0$, $w = 5$ m

If you finish early

Extra

Bronze

$x(x + 4) = 21$. Find the positive value of x.

$x = 3$

Silver

A rectangle is x cm by $(x - 3)$ cm with area 40 cm^2. Find x.

$x = 8$

Gold

A number plus its square is 42. Find the number.

$n^2 + n - 42 = 0$, $n = 6$ or $n = -7$

Platinum

A right-angled triangle has sides x, $x + 7$ and hypotenuse $x + 8$. Find x.

$x^2 - 2x - 15 = 0$, so $x = 5$

Exam question 1

4 marks

A rectangle has length $(x + 6)$ cm and width x cm. Its area is 27 cm^2. Find its perimeter.

Space for working

Mark scheme: $x^2 + 6x - 27 = 0$, $(x + 9)(x - 3) = 0$, $x = 3$. 9 cm by 3 cm, perimeter 24 cm

Objective 2 achieved

Form and solve quadratic equations from problems



Real life

Apply

A rectangle has length $(x + 5)$ m and width x m. Its area is 84 m². Find its dimensions.

$x^2 + 5x - 84 = 0 \rightarrow (x + 12)(x - 7) = 0 \rightarrow$ 7 m by 12 m

Extension: stretch yourself

Stretch

1. Solve $(x + 2)^2 = 3x + 10$

$x^2 + x - 6 = 0 \rightarrow x = 2$ or -3

2. Simplify $(x^2 - 9)/(x^2 + x - 6)$

$(x - 3)/(x - 2)$

Exit ticket: exam questions

Exit

1. Factorise $x^2 - 11x + 24$ **(2 marks)**

2. Solve $x^2 + 2x - 35 = 0$ **(3 marks)**

3. A rectangle has width x cm and length (x + 3) cm. Its area is 40 cm^2. Find x. **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 $(x - 3)(x - 8)$ [2]

Q2 $(x + 7)(x - 5)$, $x = -7$ or 5 [3]

Q3 $x^2 + 3x - 40 = 0$, $(x + 8)(x - 5) = 0$, $x = 5$ [3]

Total: ____ / 8

Today I achieved

- ✔ **Factorise and solve $x^2 + bx + c = 0$**
- ✔ **Form and solve quadratic equations from problems**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Factorise $x^2 + 9x + 20$ [2]

2. Factorise $x^2 - 64$ [1]

3. Solve $x^2 - 7x + 12 = 0$ [3]

4. The product of two consecutive even numbers is 120. Find the numbers. [3]

5. A right-angled triangle has shorter sides x cm and (x + 7) cm and hypotenuse 13 cm. Find x. [4]

Extra practice: Maths Genie → Edexcel question bank → Solving Quadratics